CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 9: **Life Processes in Plants** | Quick revision, one page

Photosynthesis

- Carbon dioxide + water $\xrightarrow{\text{sunlight, chlorophyll}}$ glucose (starch) + oxygen.
- Occurs in green parts (mainly leaves) containing chlorophyll in chloroplasts; plants are autotrophs.
- Carbon dioxide enters through stomata; water enters through root hairs; oxygen is released.
- Starch test: iodine solution turns blue-black.

Stomata and transpiration

- Stomata: tiny pores on the leaf surface, opened and closed by guard cells; exchange of gases and loss of water vapour.
- Transpiration: loss of water vapour from leaves; it cools the plant and helps in upward movement of water.

Transport and respiration

- Xylem carries water and minerals from roots to leaves; phloem carries food from leaves to other parts.
- Respiration in plants: glucose + oxygen $\rightarrow$ energy + carbon dioxide + water; takes place day and night in all living cells.

Other modes of nutrition

- Parasites: take food from a host (Cuscuta). Saprotrophs: feed on dead matter (fungi, mushroom).
- Insectivorous plants (pitcher plant) trap insects to get nitrogen. Rhizobium in root nodules of pulses fixes nitrogen.
- Fertilisers and manure supply nutrients like nitrogen to the soil.

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